

JOAO A ASCENSAO

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EDUCATION

University of California, Berkeley

2017-2023

Ph.D. in Bioengineering

Concentration in Biophysics & Computational Biology

Advisor: Oskar Hallatschek

Rice University

2012-2016

B.S. in Bioengineering

Minor in Global Health Technologies

Advisor: Oleg Igoshin

ACADEMIC POSITIONS

Harvard University

2024-present

Postdoctoral Fellow

Cambridge, MA

Advisor: Michael Desai

Universitat Pompeu Fabra

2016-2017

Fulbright Scholar & Whitaker Fellow

Barcelona, Spain

Advisor: Jordi Garcia-Ojalvo

University of California, Berkeley

Summer 2015

Amgen Scholar

Berkeley, CA

Advisor: Adam Arkin

Rice Global Health & St. Gabriel's Hospital

Summer 2014

Biomedical Devices Intern

Namitete, Malawi

The Jackson Laboratory

Summer 2013

NSF-REU Intern

Bar Harbor, ME

Advisor: Judith Blake

PUBLICATIONS & PREPRINTS

1. **Ascensao JA**, Yu Q, Hallatschek O. The evolution of genetic drift over 50,000 generations. *Genetics* (2026). doi: [10.1093/genetics/iyag199](https://doi.org/10.1093/genetics/iyag199)
2. **Ascensao JA**, Desai M. Experimental evolution in an era of molecular tools for manipulating genotypes and phenotypes. *Nature Reviews Genetics* (2026). doi: [10.1038/s41576-025-00867-6](https://doi.org/10.1038/s41576-025-00867-6)
3. **Ascensao JA**, Abedi K, Prasad A, Hallatschek O. Frequency-dependent fitness effects are ubiquitous. *bioRxiv* (2026). doi: [10.1101/2025.08.18.670924](https://doi.org/10.1101/2025.08.18.670924) [in press at *Science Advances*].
4. **Ascensao JA**, Lok K, Hallatschek O. Asynchronous abundance fluctuations can drive giant genotype frequency fluctuations. *Nature Ecology and Evolution* (2025). doi: [10.1038/s41559-024-02578-3](https://doi.org/10.1038/s41559-024-02578-3)
5. **Ascensao JA**, Denk J, Lok K, Yu Q, Wetmore KM, Hallatschek O. Rediversification following ecotype isolation reveals hidden adaptive potential. *Current Biology* (2024). doi: [10.1016/j.cub.2024.01.029](https://doi.org/10.1016/j.cub.2024.01.029)
– See [dispatch article](#) by W. Shoemaker

6. Yu Q, **Ascensao JA**, Okada T, COG-UK consortium, Boyd O, Volz E, Hallatschek O. Lineage frequency time series reveal elevated levels of genetic drift in SARS-CoV-2 transmission in England. *PLOS Pathogens* (2024). doi: [10.1371/journal.ppat.1012090](https://doi.org/10.1371/journal.ppat.1012090)
7. **Ascensao JA**, Wetmore KM, Good BH, Arkin AP, Hallatschek O. Quantifying the local adaptive landscape of a nascent bacterial community. *Nature Communications* (2023). doi: [10.1038/s41467-022-35677-5](https://doi.org/10.1038/s41467-022-35677-5)
8. **Ascensao JA***, Datta P*, Hancioglu B*, Sontag E*, Gennaro ML, Igoshin OA. Non-monotonic response to monotonic stimulus: regulation of glyoxylate shunt gene expression dynamics in *Mycobacterium tuberculosis*. *PLOS Computational Biology* (2016). doi: [10.1371/journal.pcbi.1004741](https://doi.org/10.1371/journal.pcbi.1004741)
9. **Ascensao JA***, Dolan ME*, Hill DP, Blake JA. Methodology for the inference of gene function from phenotype data. *BMC Bioinformatics* (2014). doi: [10.1186/s12859-014-0405-z](https://doi.org/10.1186/s12859-014-0405-z)

*equal contribution

AWARDS & RECOGNITIONS

2025	CFAR Scholar Award
2022	UC Berkeley International Conference Travel Grant
2020	Lloyd Scholarship (<i>departmental award</i>)
2019	Brodie Scholarship (<i>departmental award</i>)
2017-22	NSF Graduate Research Fellowship
2017-22	Berkeley Fellowship for Graduate Study
2016-17	Fulbright Scholar
2016-17	Whitaker International Program Fellowship
2016	MIT Sloan Scholar for Graduate Study [<i>declined</i>]
2016	Princeton Presidential Fellowship & Dean's Grant [<i>declined</i>]
2016	Rice Institute for Global Health–Innovator Award
2015	Outstanding Junior in Bioengineering Award
2015	Best Engineering Poster; Rice Undergraduate Research Symposium
2015	Amgen Scholar; University of California, Berkeley
2014	Global Health Award; Rice Undergraduate Research Symposium
2012-16	President's Honor Roll

TALKS

2026	Condensed Matter Theory Seminar, Harvard University	Cambridge, MA
2024	Society of Industrial & Applied Math, Life Sciences Conference	Portland, OR
2023	Molecular Mechanisms in Evolution, Gordon Conference	Easton, MA
2023	Seminar at Max Planck Institute for the Science of Light	Erlangen, Germany
2022	Bay Area Population Genetics	Berkeley, CA
2022	Evolutionary Dynamics and Processes	Plön, Germany
2020	Microbial Ecology and Evolution Virtual (MEEVirtual)	Virtual
2019	West Coast Bacterial Physiology	Asilomar, CA
2019	UC Berkeley Bioengineering Retreat (Brodie Scholar Talk)	Santa Cruz, CA
2017	Whitaker Fellows Seminar	Lisbon, Portugal
2013	Gene Ontology Consortium Meeting	Bar Harbor, ME

POSTERS

2026	Function of Evolving Systems, Gordon Conference	Waterville Valley, NH
2025	Microbial Population Biology, Gordon Conference	Andover, NH
2024	The Allied Genetic Conference (TAGC)	National Harbor, MD
2023	Bay Area Population Genetics	Palo Alto, CA
2023	Les Houches Theoretical Biological Physics	Les Houches, France
2023	Molecular Mechanisms in Evolution, Gordon Conference	Easton, MA
2023	Stochastic Physics in Biology, Gordon Conference	Ventura, CA
2023	Berkeley Statistical Mechanics Meeting	Berkeley, CA
2019	Microbial Population Biology, Gordon Conference	Andover, NH
2015	Bioengineering Undergraduate Poster Symposium	Houston, TX
2015	UC Berkeley Amgen Scholars Poster Session	Berkeley, CA
2015	Rice Undergraduate Research Symposium	Houston, TX
2014	Bioengineering Undergraduate Poster Symposium	Houston, TX

RESEARCH SUPERVISION

2024-present	Alex Poret, graduate student, biophysics
2024-present	Jenya Belousova, graduate student, evolutionary biology
2023-2025	Keon Abedi, undergraduate student, physics
spring 2024	Kira Buttrey, rotation student, bioengineering
fall 2023	Sarah Wasinger, rotation student, bioengineering
fall 2023	Aaron Fultineer, rotation student, physics
2020-2023	Kristen Lok, undergraduate student, bioengineering
2020-2021	Can Goksal, undergraduate student (co-supervised with QinQin Yu), biology
2019-2020	Nicole Tin, undergraduate student, bioengineering

GRADUATE COURSEWORK

2017	Statistical Mechanics I
2019	Statistical Mechanics II
2017	Computational Molecular Biology
2018	Computational Statistics
2018	Genomics for Microbiology
2018	Mathematical Models of Infectious Disease
2019	Probabilistic Models & Machine Learning in Genomics
2018	Microbial Diversity & Evolution
2018	Biological Regulatory Mechanisms
2024	Immunology

ADDITIONAL EDUCATION

Les Houches School of Physics Theoretical Biological Physics	Summer 2023
Marine Biological Laboratory Microbial Diversity	Summer 2019

SERVICE & OTHER WORK

“LS50: Integrated Science”, Harvard University Instructor	2025-2026 Cambridge, MA
· Designed a lab class where student groups will tackle a novel research problem—exploring HIV bnAb evolution with high-throughput antibody libraries and statistical analysis.	

- Projects were designed to minimize risk, such that they are appropriate for undergraduates.

Tutoring Plus

Volunteer tutor

2024-2025

Cambridge, MA

- Volunteered with local non-profit to provide free tutoring services to K-12 students.

“Data Science for Biology”, UC Berkeley

Graduate Student Instructor

2022

Berkeley, CA

- Prepared and gave lectures for class discussion section.
- Held office hours for students.
- Helped to design coding assignments and graded them.

“Biocomputing”, Universitat Pompeu Fabra

Teaching Assistant

2017

Barcelona, Spain

- Helped second year Bioengineering undergraduate students with coding assignments (in Spanish).
- Prepared and gave lectures for class discussion section.

Virginia Science Olympiad

Volunteer

2011-2019

Northern Virginia and remotely

- Writer for regional and state Science Olympiad forensics events for Middle School students.
- Continue to work remotely since moving away from Virginia.

Fulbright Outreach

Volunteer

2016-2017

Barcelona, Spain

- Gave talks to high school students in Barcelona about topics in modern biology research, and my own research projects.

Brown College, Rice University

Academic Fellow

2016-2017

Houston, TX

- Tutored individual college members in a variety of subjects, including physics, math and computational engineering.
- Led review sessions before exams for freshman physics & math classes.

Biomedical Engineering Society

Mentor

2014-2016

Houston, TX

- Served as a year-long resource and mentor for underclassman bioengineers.
- Advised mentees on class selection, internships, research, etc.

“Bioengineering Fundamentals”, Rice University

Teaching Assistant & Grader

2014

Houston, TX

- Graded homework and led homework sessions for introductory bioengineering class.

Updated: September 21, 2026